MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 01** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 20**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. The ratio of the length of a vertical rod and the length of its shadow on the ground is $1 : \sqrt{3}$. The angle of elevation of the sun is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Let vertical rod height $= x$ and shadow length $= \sqrt{3}x$. Then $\tan\theta = \dfrac{x}{\sqrt{3}x} = \dfrac{1}{\sqrt{3}} = \tan 30° \implies \theta = \boxed{30°}$.

Q2. If the length of the shadow of a tower on the level ground is $\sqrt{3}$ times the height of the tower, then the angle of elevation of the sun is: [1]

(A) $45°$ (B) $30°$ (C) $60°$ (D) $90°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: Let tower height $= h$, so shadow length $= \sqrt{3}h$. Hence $\tan\theta = \dfrac{h}{\sqrt{3}h} = \dfrac{1}{\sqrt{3}} \implies \theta = \boxed{30°}$.

Q3. At a certain time of the day, if the length of the shadow of a vertical pole is equal to its height, then the sun's altitude is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: Since height = shadow length ($h = s$), we have $\tan\theta = \dfrac{h}{s} = 1 = \tan 45° \implies \theta = \boxed{45°}$.

Q4. A vertical pole 6 m high casts a shadow $2\sqrt{3}$ m long on the level ground. The sun's elevation is: [1]

(A) $60°$ (B) $45°$ (C) $30°$ (D) $90°$

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Justification: $\tan\theta = \dfrac{\text{Height}}{\text{Shadow}} = \dfrac{6}{2\sqrt{3}} = \dfrac{3}{\sqrt{3}} = \sqrt{3} = \tan 60° \implies \theta = \boxed{60°}$.

Q5. Assertion (A): If the length of the shadow of a vertical pole is equal to its height, then the angle of elevation of the sun is $45°$.
Reason (R): For any right-angled triangle, when the opposite side equals the adjacent side, $\tan\theta = 1$, which gives $\theta = 45°$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

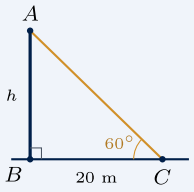
Justification: Both Assertion (A) and Reason (R) are mathematically true, and $\tan\theta = \dfrac{\text{Opposite}}{\text{Adjacent}} = 1 \implies \theta = 45°$ directly explains why equal pole height and shadow give sun's elevation $45°$.

SECTION B: Short Answer Questions — MATHS MANTHAN | (11 Marks)

Q6. A vertical pole stands on level ground. From a point on the ground 20 m away from the foot of the pole, the angle of elevation of the top is $60°$. Find the height of the pole. [2]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Let $AB = h$ m be the vertical pole and C be the point on the ground 20 m from its foot B, with $\angle ACB = 60°$. [0.5 Mark]
In right-angled $\triangle ABC$ ($\angle B = 90°$):



$$\tan 60° = \frac{AB}{BC} \implies \sqrt{3} = \frac{h}{20}$$

Applying the trigonometric ratio gives $h = 20\sqrt{3}$ m. [1 Mark]
Substituting $\sqrt{3} = 1.732$, we get $h = 20 \times 1.732 = 34.64$ m.

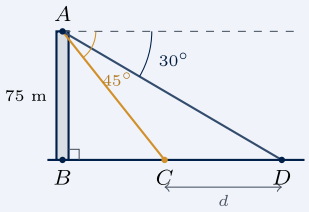
$$\therefore \text{ Height of pole} = \boxed{20\sqrt{3} \text{ m} \approx 34.64 \text{ m}}$$

Final evaluated result: [0.5 Mark]

Q7. As observed from the top of a 75 m high lighthouse from the sea level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. [Take $\sqrt{3} = 1.732$] **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let $AB = 75$ m be the lighthouse and C, D be the two ships on the same side with angles of depression 45° and 30° respectively. By alternate interior angles, $\angle ACB = 45°$ and $\angle ADB = 30°$. [0.5 Mark]



In right $\triangle ABC$:

$$\tan 45° = \frac{AB}{BC} \implies 1 = \frac{75}{BC} \implies BC = 75 \text{ m}$$

Obtaining $BC = 75$ m: [1 Mark]

In right $\triangle ABD$:

$$\tan 30° = \frac{AB}{BD} \implies \frac{1}{\sqrt{3}} = \frac{75}{BD} \implies BD = 75\sqrt{3} \text{ m}$$

Obtaining $BD = 75\sqrt{3}$ m: [1 Mark]

Distance between the two ships $= CD = BD - BC = 75\sqrt{3} - 75 = 75(\sqrt{3} - 1)$ m.

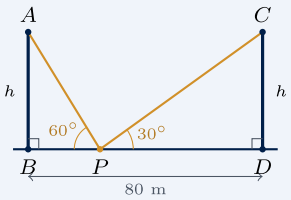
$$CD = 75(1.732 - 1) = 75 \times 0.732 = \boxed{54.9 \text{ m}}$$

Final distance between the ships: [0.5 Mark]

Q8. Two poles of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30° respectively. Find the height of the poles and the distances of the point from the poles. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let $AB = CD = h$ m be the two equal poles across road $BD = 80$ m, and P be the point on BD such that $BP = x$ m and $PD = (80 - x)$ m, with $\angle APB = 60°$ and $\angle CPD = 30°$. [0.5 Mark]



In right $\triangle ABP$:

$$\tan 60° = \frac{AB}{BP} \implies h = \sqrt{3}\,x \qquad \text{— (i)}$$

In right $\triangle CDP$:

$$\tan 30° = \frac{CD}{PD} \implies \frac{1}{\sqrt{3}} = \frac{h}{80 - x} \implies h = \frac{80 - x}{\sqrt{3}} \qquad \text{— (ii)}$$

Forming both equations: [1 Mark]

Equating (i) and (ii): $\sqrt{3}\,x = \dfrac{80 - x}{\sqrt{3}} \implies 3x = 80 - x \implies 4x = 80 \implies x = 20$ m, and $PD = 80 - 20 = 60$ m. [1 Mark]

Substituting $x = 20$ in (i): $h = 20\sqrt{3} = 20 \times 1.732 = 34.64$ m.

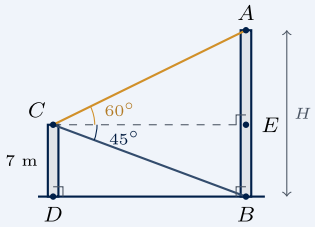
$$\therefore \text{ Height } = \boxed{20\sqrt{3} \text{ m } \approx 34.64 \text{ m}}, \qquad \text{Distances } = \boxed{20 \text{ m and } 60 \text{ m}}$$

Finding both height and distances: [0.5 Mark]

Q9. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Determine the height of the tower. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let building $CD = 7$ m and cable tower $AB = H$ m. Draw $CE \perp AB$ so $BE = CD = 7$ m and $CE = DB$. Here $\angle ACE = 60°$ and $\angle ECB = \angle CBD = 45°$. [0.5 Mark]



In right $\triangle CDB$:

$$\tan 45° = \frac{CD}{DB} \implies 1 = \frac{7}{DB} \implies DB = CE = 7 \text{ m}$$

Finding horizontal distance CE: [1 Mark]

In right $\triangle AEC$:

$$\tan 60° = \frac{AE}{CE} \implies \sqrt{3} = \frac{AE}{7} \implies AE = 7\sqrt{3} \text{ m}$$

Finding upper segment AE: [1 Mark]

Total height of the cable tower $AB = AE + EB = 7\sqrt{3} + 7 = 7(\sqrt{3} + 1)$ m.

$$AB = 7(1.732 + 1) = 7 \times 2.732 = \boxed{7(\sqrt{3} + 1) \text{ m} \approx 19.124 \text{ m}}$$

Total height calculation: [0.5 Mark]

SECTION C: Applied Case Study

MATHS MANTHAN | $(1 \times 4 = 4$ Marks)

Q10. CASE STUDY: The Historic India Gate Monument (New Delhi)

India Gate stands at a height of 42 m. A student at point P on the ground measures the angle of elevation of the top (A) as 60°. Another student stands further away at point Q in the same vertical plane and observes the top at an angle of elevation of 30°. The foot of the monument is B.

(i) At 12 noon, when the sun's rays are inclined at 45° to the horizontal ground, calculate the length of the shadow cast by India Gate. **[1]**

(ii) Find the distance of student P from the foot of the monument B. **[1]**

(iii) Find the distance between the two students (i.e., length PQ). **[2]**

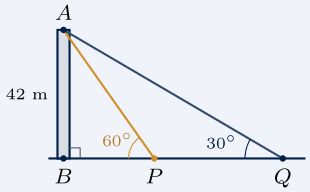
OR

Find the ratio of the length of the shadow cast by India Gate when the sun's altitude is 30° to that when the sun's altitude is 60°. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) When the sun's rays are inclined at $\theta = 45°$:

$$\tan 45° = \frac{AB}{\text{Shadow}} \implies 1 = \frac{42}{\text{Shadow}} \implies \text{Shadow} = \boxed{42 \text{ m}}$$



Subpart (i): [1 Mark]

(ii) In right $\triangle ABP$ ($\angle APB = 60°$):

$$\tan 60° = \frac{42}{BP} \implies BP = \frac{42}{\sqrt{3}}$$

$$\implies BP = \boxed{14\sqrt{3} \text{ m} \approx 24.25 \text{ m}}$$

Subpart (ii): [1 Mark]

(iii) Main Choice: In right $\triangle ABQ$ ($\angle AQB = 30°$), $\tan 30° = \dfrac{AB}{BQ} \implies BQ = 42\sqrt{3}$ m. [1 Mark]

Hence $PQ = BQ - BP = 42\sqrt{3} - 14\sqrt{3} = 28\sqrt{3}$ m.

$$PQ = 28 \times 1.732 = \boxed{28\sqrt{3} \text{ m} \approx 48.50 \text{ m}}$$

Subpart (iii) distance PQ: [1 Mark]

(iii) OR Choice: At sun altitude $30°$, shadow $s_{30} = 42 \cot 30° = 42\sqrt{3}$ m. At altitude $60°$, shadow $s_{60} = 42 \cot 60° = \dfrac{42}{\sqrt{3}}$ m. [1 Mark]

$$\text{Required Ratio} = \frac{s_{30}}{s_{60}} = \frac{42\sqrt{3}}{42/\sqrt{3}} = \sqrt{3} \times \sqrt{3} = \boxed{3 : 1}$$

Final ratio: [1 Mark]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★